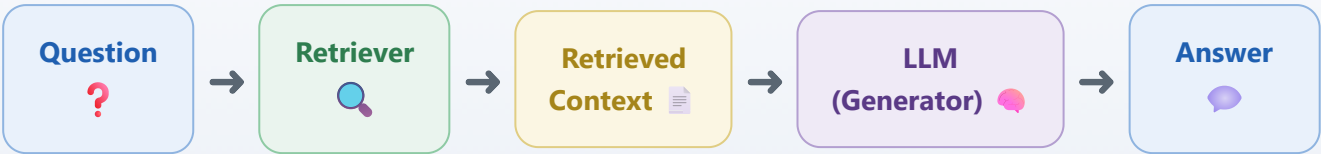


Complete RAG Evaluation Guide

Evaluate RAG at three levels: 1) Retrieval Quality 2) Generation Quality 3) End-to-End & Production



RAG Evaluation at a Glance

Retrieval Evaluation

Did we retrieve the right information?

Metrics: Precision, Recall, Hit@K, MRR, NDCG, Context Precision, Context Recall.

Generation Evaluation

Did the LLM use the information correctly?

Metrics: Faithfulness (Groundedness), Answer Relevance, Answer Correctness, Hallucination Rate.

End-to-End & Production

Overall system performance & user value.

Metrics: Quality + Latency + Cost + Reliability + User Feedback.

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- 1. Retrieval Evaluation Metrics
- 2. Generation Evaluation Metrics
- 3. End-to-End & Production Metrics
- 4. Evaluation Dataset
- 5. LLM-as-a-Judge Evaluation
- 6. Complete RAG Evaluation Architecture
- 7. Popular RAG Evaluation Tools

1. Retrieval Evaluation Metrics

1.1 Precision@K

TOP-K PRECISION

Of the retrieved chunks, how many are relevant?

$$\text{Precision@K} = \frac{\# \text{ Relevant Retrieved}}{K}$$

Example (K=5)

Rank	Retrieved Chunk	Relevant?
1	Chunk 1	✓
2	Chunk 2	✗
3	Chunk 3	✓
4	Chunk 4	✗
5	Chunk 5	✓

Relevant = 3 → Precision@5 = 3/5 = **0.60 (60%)**

1.2 Recall@K

Of all relevant chunks, how many were retrieved?

$$\text{Recall@K} = \frac{\# \text{ Relevant Retrieved}}{\# \text{ Total Relevant}}$$

Example (K=5)

Total relevant chunks in DB = 4
Retrieved relevant chunks = 3
Recall@5 = 3/4 = **0.75 (75%)**

Precision vs Recall

Precision → Of what I retrieved, how much was relevant?
Recall → Of everything relevant, how much did I retrieve?

1.3 Hit Rate / Hit@K

Was at least one relevant document retrieved in top K?

$$\text{Hit@K} = \{ 1, \text{ if } \geq 1 \text{ relevant in top K} \quad 0, \text{ otherwise } \}$$

Example (K=5)

Top 5: ✗ ✗ ✓ ✗ ✗
Hit@5 = 1

If evaluating 100 questions and 85 have ≥ 1 relevant in top 5:
Hit@5 = 85/100 = **0.85 (85%)**

1.4 MRR

MEAN RECIPROCAL RANK

How high is the first relevant result?

For each query: $RR = 1 / \text{rank of first relevant document}$

Example

Query	Rank of First Relevant	RR
Q1	1	1/1 = 1.00
Q2	2	1/2 = 0.50
Q3	4	1/4 = 0.25

MRR = $(1 + 0.50 + 0.25) / 3 = \mathbf{0.583 \text{ (58.3\%)}}$
MRR only cares about the first relevant result.

1.5 NDCG@K

NORMALIZED DISCOUNTED CUMULATIVE GAIN

Considers relevance and ranking position. Handles graded relevance.

Relevance Levels Example

- 3 = Highly Relevant
- 2 = Relevant
- 1 = Slightly Relevant
- 0 = Irrelevant

Example (K=4)

Rank	Relevance
1	3
2	2
3	0
4	1

$DCG@K = \sum_{i=1..K} (2^{rel_i} - 1) / \log_2(i + 1)$

$IDCG@K = DCG@K \text{ of ideal ranking (sorted by relevance desc)}$

$NDCG@K = DCG@K / IDCG@K \quad (\text{Range: } 0 \text{ to } 1)$

1.0 = perfect ranking, 0 = worst

1.6 Context Precision

RAG SPECIFIC

How relevant is the retrieved context to the question?

$$\text{Context Precision} = \frac{\# \text{ Relevant Retrieved Chunks}}{\# \text{ Retrieved Chunks}}$$

Example

Retrieved 5 chunks. Relevant = 3
Context Precision = $3/5 = \mathbf{0.60 \text{ (60\%)}}$

1.7 Context Recall

RAG SPECIFIC

Did retrieval provide all the information needed to answer?

Context Recall =

Relevant / Required Info Retrieved

Total Required Info

Example

Answer needs 4 pieces of information.
Retrieved 3 of them.
Context Recall = 3/4 = **0.75 (75%)**

2. Generation Evaluation Metrics

2.1 Faithfulness / Groundedness

Is the answer supported by the retrieved context? Break answer into claims.

Example

Context:

- Refunds within 30 days.
- Original receipt required.

LLM Answer (3 claims):

1. Refunds within 30 days.	✓
2. Original receipt required.	✓
3. Refunds available for 60 days.	✗

Faithfulness = $\frac{\text{Supported Claims}}{\text{Total Claims}}$ = 2/3 = 0.667 (66.7%)

2.2 Answer Relevance

Does the answer actually address the question? Usually evaluated using semantic similarity or LLM judge.

Q: "What is the refund period?"

Answer A: "The refund period is 30 days."

Score: 0.95 (Highly relevant)

Answer B: "The company was founded in 2015."

Score: 0.10 (Not relevant)

2.3 Answer Correctness

Is the generated answer factually correct compared to ground truth?

Ground Truth: "Refunds are available within 30 days."

Generated: "Customers can request a refund within 30 days."

→ Meaning matches. High correctness.

(Score via semantic/factual comparison or LLM judge)

2.4 Hallucination Rate

Measures unsupported claims in the answer.

Example

Total claims = 10
Unsupported claims = 2

Hallucination Rate = 2/10 = 0.20 (20%)

Grounded Claim Rate = 1 – 0.20 = 0.80 (80%)

3. End-to-End & Production Metrics

🕒 Latency

Measure each stage:

- Retrieval latency
- Reranking latency
- Generation latency
- Total latency

Latency = T_{response} - T_{request}

Example

10:00:00.000 → request
10:00:01.800 → response
Latency = **1.8 seconds**

💰 Cost

Usually based on LLM token usage.

Cost = (Input Tokens × Input Price) + (Output Tokens × Output Price)

- Input tokens
- Output tokens
- Total tokens

Example

Input = 2,000 tokens
Output = 500 tokens
Input price = \$X / 1M tokens
Output price = \$Y / 1M tokens
Cost = 2000·X/1M + 500·Y/1M

📄 Token Usage

Track tokens for performance & cost.

- Input tokens
- Output tokens
- Total tokens

🛡️ Reliability

- Retrieval failure rate
- Empty retrieval rate
- Timeout rate
- Error rate
- API failure rate

🏆 Quality (Production)

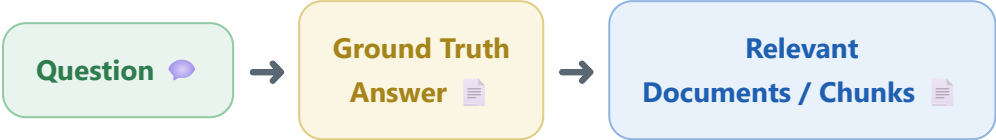
- Faithfulness score
- Answer correctness
- Hallucination rate
- "No answer" accuracy
- User feedback score

⚙️ **System**

- Throughput (QPS)
- Resource usage (CPU, Memory)
- Vector DB latency
- Indexing health
- Cache hit rate

4. Evaluation Dataset

Create a high-quality evaluation dataset.



Example Table

Question	Ground Truth Answer	Relevant Chunk(s)
What is refund period?	30 days	chunk_17
How can I reset password?	Use reset link	chunk_42
What is annual fee?	\$99	chunk_83
What is late fee?	\$10 per day	chunk_91
Do you offer free trial?	Yes, 7 days	chunk_15

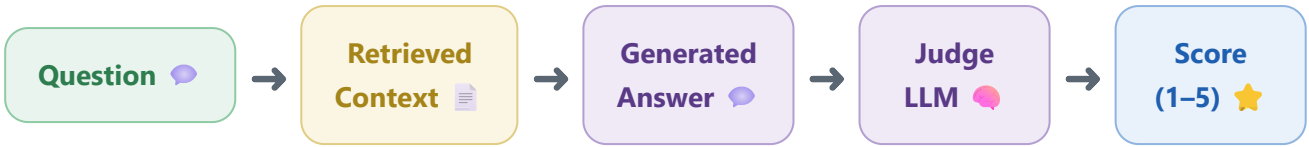
Dataset Size Recommendation

- 50–100 questions → basic testing
- 200–500 questions → strong evaluation
- 500+ questions → production-scale evaluation

Include diverse question types

- Simple factual · Multi-hop · Comparison · Numerical
- Long-context · Ambiguous · No-answer

5. LLM-as-a-Judge Evaluation



Judge Criteria (Example)

- 1. Is the answer relevant?
- 2. Is it factually correct?
- 3. Is it supported by context?
- 4. Does it contain hallucinations?

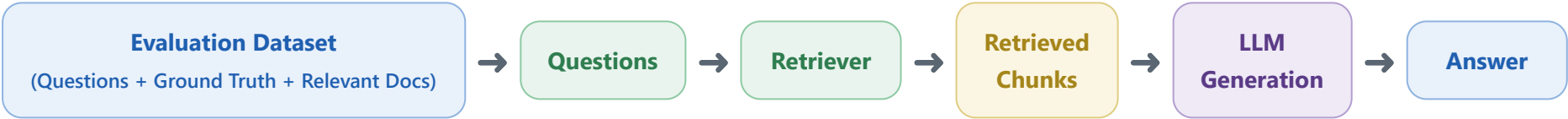
Example Score

Faithfulness	4 / 5
Answer Relevance	5 / 5
Correctness	4 / 5
Hallucination	1 / 5
Overall Score	4.3 / 5

Best Practice

Use LLM-as-a-Judge + Automated Metrics + Human Evaluation for reliable results.

6. Complete RAG Evaluation Architecture



Retrieval Evaluation

- Precision@K · Recall@K
- Hit@K · MRR
- NDCG · Context Precision
- Context Recall

Generation Evaluation

- Faithfulness
- Answer Relevance
- Answer Correctness
- Hallucination Rate

Overall RAG Score

Production Monitoring & Continuous Improvement

7. Popular RAG Evaluation Tools

Ragas

Open-source framework for RAG evaluation. Metrics: Context Precision, Context Recall, Faithfulness, Answer Relevance, Correctness, Hallucination, etc.

DeepEval

Evaluate LLMs, RAG, Agents, and conversational systems with many ready-to-use metrics and test cases.

LangSmith

Tracing, evaluation, dataset management, experiment comparison, and observability.

TruLens

Evaluate LLM applications with feedback functions, groundedness, relevance, and more.

Arize Phoenix

Production evaluation, monitoring, prompt tracing, drift detection, and more.

Quick Reference — Metrics Cheat Sheet

Retrieval Metrics

- **Precision@K** → Relevant retrieved / K
- **Recall@K** → Relevant retrieved / Total relevant
- **Hit@K** → 1 if ≥ 1 relevant in top K, else 0
- **MRR** → 1 / rank of first relevant
- **NDCG** → DCG / IDCG (considers relevance + rank)
- **Context Precision** → Relevance of retrieved context
- **Context Recall** → Completeness of retrieved context

Generation Metrics

- **Faithfulness** → Supported claims / Total claims
- **Answer Relevance** → Does answer address question?
- **Answer Correctness** → Matches ground truth answer
- **Hallucination Rate** → Unsupported claims / Total claims

 **Retrieval Metrics** tell you whether you found the **right information**.

VS

 **Generation Metrics** tell you whether the LLM produced the **right answer from that information**.